RAD-SMART | Radiotherapy scheduling

Health-a-thon 2026

About the Team / Organisation

A. The team. Radiotherapy department, Kasturba Medical College, Manipal (MAHE), with two engineers: Abhinand T M (technical lead) and P S Raghotham Rao (lead engineer: the scheduling engine, digital twin, CARE plugin and Sarvam AI layer). Built on CARE, the open-source hospital information system of the Open Healthcare Network. Team code YU8PAK.

B. The doctors. Dr Akshay Dinesan (team leader, doctor partner) wrote the problem statement from daily work, brought the department's records and owns the clinical rules and the pilot. Dr Shirley Lewis Salins and Dr Umesh Velu reviewed the rules and the design. The department's technologists and doctors answered our questions in writing on 21 September and 24 October; those answers are now rules in the code.

Problem Statement (use case specific)

One linear accelerator (Elekta Versa HD) treats **70 to 90 patients a day**, 08:30 until about 01:00. A course means coming every weekday for one to seven weeks. Times are given by hand, without knowing how long a session takes, so the delay builds through the day.

- In the department's records (55 days, 3,137 visits, anonymised totals), **one patient in ten waits more than 109 minutes**, and **22% are treated over an hour late**.
- A breakdown is handled by phone; patients already travelling are not told.
- 69% of India's radiotherapy facilities have one machine: this is the common case.

Users: radiation oncologists, technologists, reception, patients and families, the head of department.

AI Solution Description

RAD-SMART plans the machine's day so that waits are short and every department rule is kept. It runs **inside CARE** as a plugin. The doctor enters a course on the patient's visit and approves a first day and **one daily time** for the whole course. The technologist runs the day on a live board, and when the machine breaks down RAD-SMART **re-plans the rest of the day in seconds** under the department's own rules. Patients get each time on a private phone page and WhatsApp, **in their own language with a voice note**. A person approves every change; RAD-SMART decides only when, never whether or how.

Technology and Tools Used

- **Gradient boosting** (scikit-learn) predicts each session's minutes.
- **Mixed-integer optimisation** (SciPy with HiGHS) plans with the rules as hard constraints; an independent checker re-verifies every plan; a calibrated **digital twin** tests plans first.
- **Sarvam AI**: Translate, Bulbul v3 voice notes, Sarvam-105B plain-language explanations, each output checked for changed times.
- Django and React plugins in CARE, PostgreSQL, one Docker image on Microsoft Azure.

Key Features of the solution

- **Standing daily time** per course, inside the patient's preferred morning, afternoon or evening; better times offered as slots free up, agreed by staff with the patient.
- **Day board** with Call next, out of turn with a reason, and **Priority** and same-day **emergency starts** that staff decide.
- **Breakdown re-planning** in about 10 to 14 seconds, approved by a person.
- **Messages** in Kannada, Malayalam, Hindi and Konkani with voice notes; a phone page with one-tap replies and QR check-in.
- **Insights**: measured waits against the 109-minute baseline.

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Build Sprint Progress

From a Round 1 engine and simulation (19 Sep) to a working application inside CARE, live on Azure:

- **24 Oct**: the department's rules (standing times, new starts by 17:00, the 60-minute breakdown rule, roles) built into the engine and a CARE app.
- **25 Oct**: Sarvam messages and voice notes, the patient phone page, QR check-in, explanations.
- **26 Oct**: deployed on Microsoft Azure in one Docker image, open to try.
- **27 Oct**: from the doctor partner, preferred part of the day and a live queue for early and late patients.
- **3 Nov**: Priority override, same-day emergency starts, a guided demonstration.

Not yet: automatic WhatsApp sending, spoken replies, the oncology-system feed, two machines.

User Validation and AI Evaluation

- On the department's 55 recorded days, re-planned in a calibrated twin: **90th-percentile wait 108.6 to 70.4 min (−35%)**; over-an-hour lateness **22.2% to 5.7%**.
- After a 150-minute breakdown: median wait **104 to 39 min**, new plan in 14 s. No hard rule broken.
- Prediction error 2.15 min; optimum in 7.1 s; **112 of 112** translations kept every time; 20 of 20 explanations with no invented time; 49 automated tests pass.
- The department's feedback changed the design (standing times, breakdown rules, preferred times, out-of-turn). Limits: a replay, not a trial; native-speaker ratings and a technologist trial next.

Safety and Responsible AI

- **People decide**: a doctor approves each course; a technologist approves every change on the day; staff can always override.
- **Rules are guaranteed**: hard constraints in the optimiser, re-checked independently.
- **AI fenced in**: language models never schedule; a translation or explanation that changes a time is discarded; without Sarvam it works in English.
- **Privacy**: no names or numbers sent to Sarvam; first name only on the patient page; synthetic demonstration data; DPDP Act consent and ethics review for the pilot.

Deployment and Sustainability Plan

- Live on Azure as one Docker image; the pilot runs the same image on the hospital's server or in an Indian Azure region. Installs as a CARE plugin, no change to CARE.
- Costs: hardware ₹0 to 2.25 lakh (existing PCs cost nothing); about ₹1 a message, ₹3 to 5 a course; open source, no licence fee.
- 90-day pilot on the Versa HD (baseline, shadow, live; target −25%), then the second machine and other CARE hospitals.

Supporting Links

- **MVP, no password**: radsmart.braveglacier-5f9807f7.uaenorth.azurecontainerapps.io
- **Video demo walkthrough**: psraghothamrao8.github.io/RAD-SMART/#demo
- **Website and documents**: psraghothamrao8.github.io/RAD-SMART
- **Code**: github.com/psraghothamrao8/RAD-SMART